

RecycleVision

AI-Powered Recycling Intelligence System

Tanniru Varun

May 16, 2026

GitHub Repository:

<https://github.com/NaniT55/recyclevision-ai>

Frontend Deployment:

<https://recyclevision-ai.vercel.app/>

Backend Deployment:

<https://recyclevision-backend-olwz.onrender.com>

1 Problem Statement

Recycling facilities receive large batches of mixed plastic waste materials daily. Manual inspection and classification of these materials is time-consuming, inconsistent, and operationally expensive. Facilities require an intelligent system capable of analyzing incoming recycling batches and generating operational insights such as bottle count estimation, PET composition analysis, contamination risk detection, quality grading, and recycling value estimation.

The objective of this project is to design and implement a working AI-powered prototype capable of analyzing recycling bottle images and generating structured operational intelligence through a modern web-based dashboard.

2 Project Overview

RecycleVision is a full-stack AI-powered recycling intelligence platform that uses multi-modal vision-language models to analyze recycling bottle batches.

The platform performs:

- Bottle count estimation
- PET vs Non-PET composition estimation
- Contamination risk assessment
- Confidence scoring
- Quality grading
- Estimated recycling value prediction
- Structured JSON report generation

The system includes:

- React + Vite frontend dashboard
- FastAPI backend API
- Groq Vision AI integration
- SQLite/local storage report management
- Deployed cloud infrastructure

3 System Architecture

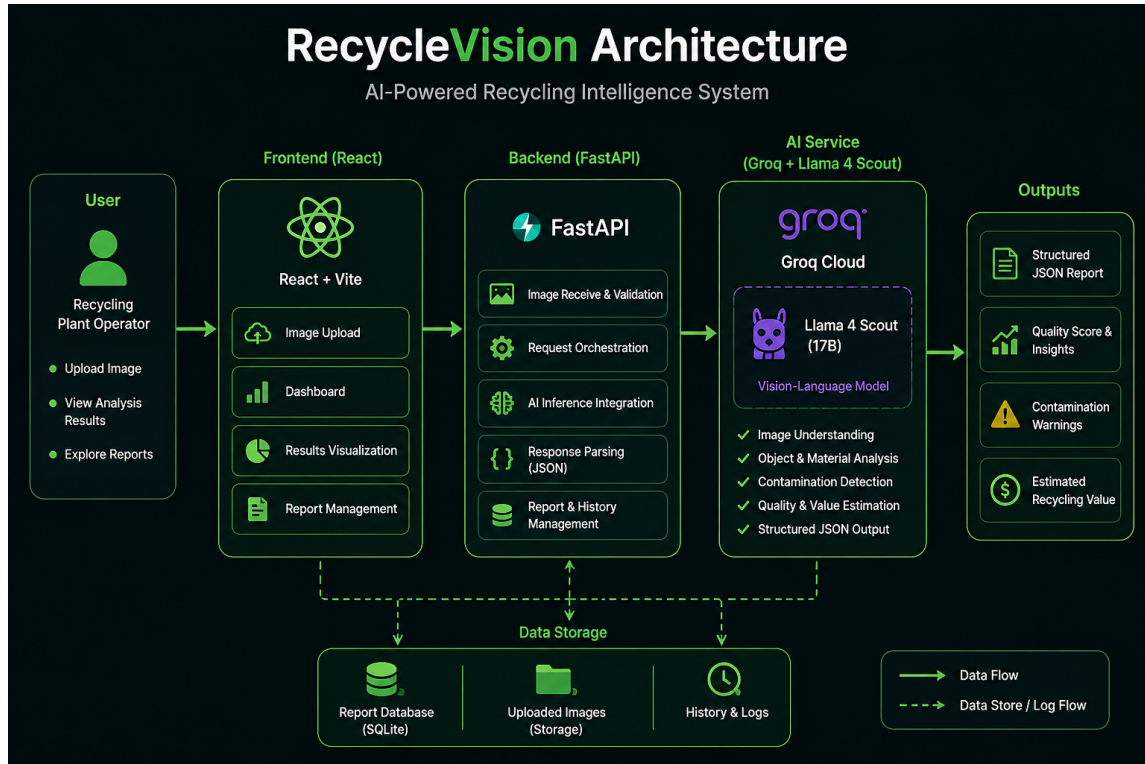


Figure 1: RecycleVision System Architecture

3.1 Architecture Flow

1. User uploads recycling image through React frontend.
2. Frontend sends image to FastAPI backend.
3. Backend processes image and forwards request to Groq Vision API.
4. Llama 4 Scout multimodal model analyzes recycling content.
5. AI response is converted into structured JSON.
6. Backend returns operational insights.
7. Frontend dashboard displays results and stores reports.

3.2 Technology Stack

Component	Technology
Frontend	React, Vite, CSS
Backend	FastAPI, Python
AI/ML	Groq API, Llama 4 Scout Vision Model
Database	SQLite / Browser Local Storage

Deployment	Vercel + Render
Version Control	Git + GitHub

4 AI/ML Approach

4.1 Chosen Approach

The prototype uses a pretrained multimodal Vision-Language Model (VLM) through the Groq API. Specifically, the project integrates Meta’s Llama 4 Scout multimodal model.

The model receives recycling bottle images and performs semantic reasoning to generate structured recycling intelligence.

4.2 Why This Approach Was Selected

The following reasons motivated the decision:

- Rapid prototyping without training custom models
- No labeled dataset required initially
- Strong multimodal reasoning capabilities
- Flexible interpretation of complex recycling scenes
- Faster implementation within limited development time
- Easier deployment and infrastructure management

4.3 Alternative Approaches Considered

4.3.1 YOLO Object Detection

YOLO-based object detection was considered for accurate bottle counting and material detection.

Rejected Because:

- Requires labeled training dataset
- Requires training and hyperparameter tuning
- Increased implementation complexity
- Limited time for dataset preparation

4.3.2 Image Segmentation Models

Segmentation-based approaches such as Mask R-CNN were considered.

Rejected Because:

- Computationally expensive
- Requires detailed pixel-level annotations
- Beyond prototype scope

4.3.3 OCR Pipelines

OCR-based label extraction was considered.

Rejected Because:

- Not essential for prototype functionality
- Label visibility varies significantly
- Additional preprocessing complexity

4.4 Trade-Offs

The chosen VLM-based approach provides flexibility and rapid development at the cost of exact object detection precision.

Advantages:

- Faster development
- Strong semantic reasoning
- No training pipeline required

Limitations:

- Approximate bottle counting
- Lower geometric precision than detection models
- Results depend on model reasoning quality

5 Application Features

5.1 Implemented Features

- Recycling image upload
- AI-powered recycling analysis
- Bottle count estimation
- PET composition estimation
- Contamination analysis
- Confidence scoring
- Batch grading
- Estimated recycling value generation
- JSON report generation
- Reports dashboard
- Architecture visualization popup
- Responsive UI

5.2 Homepage

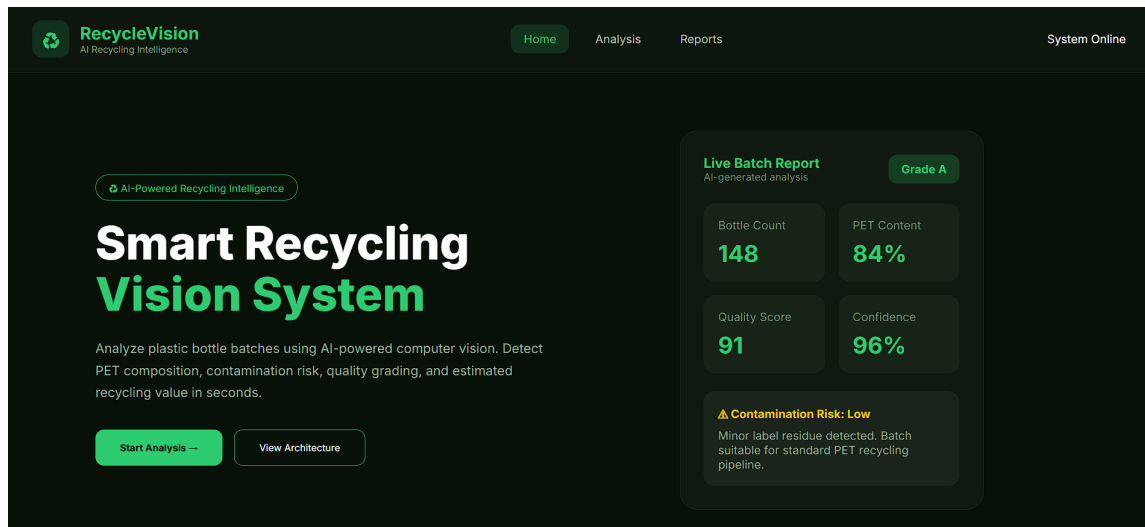


Figure 2: RecycleVision Homepage

5.3 Analysis Dashboard

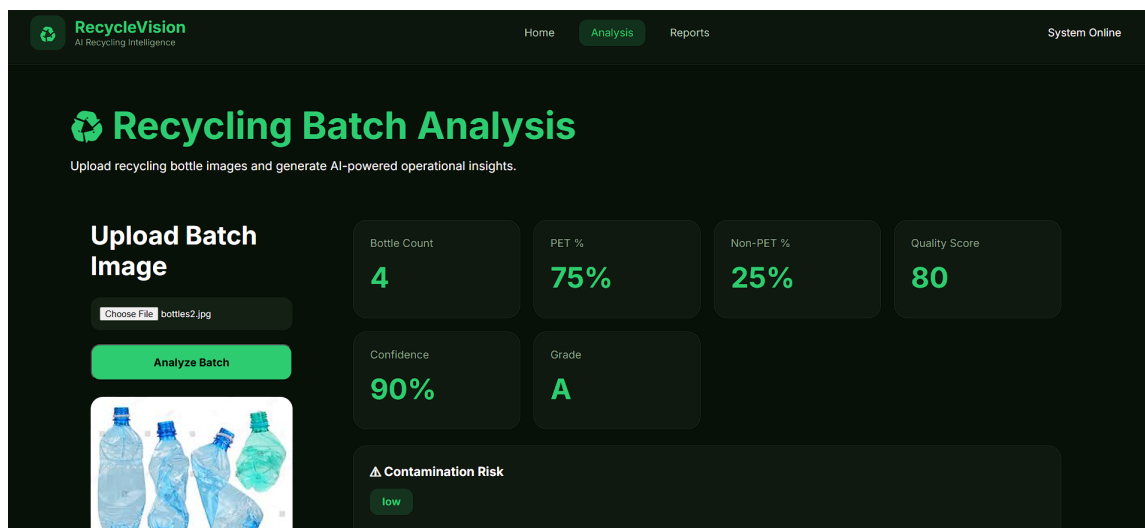


Figure 3: AI Analysis Dashboard

5.4 Reports Dashboard

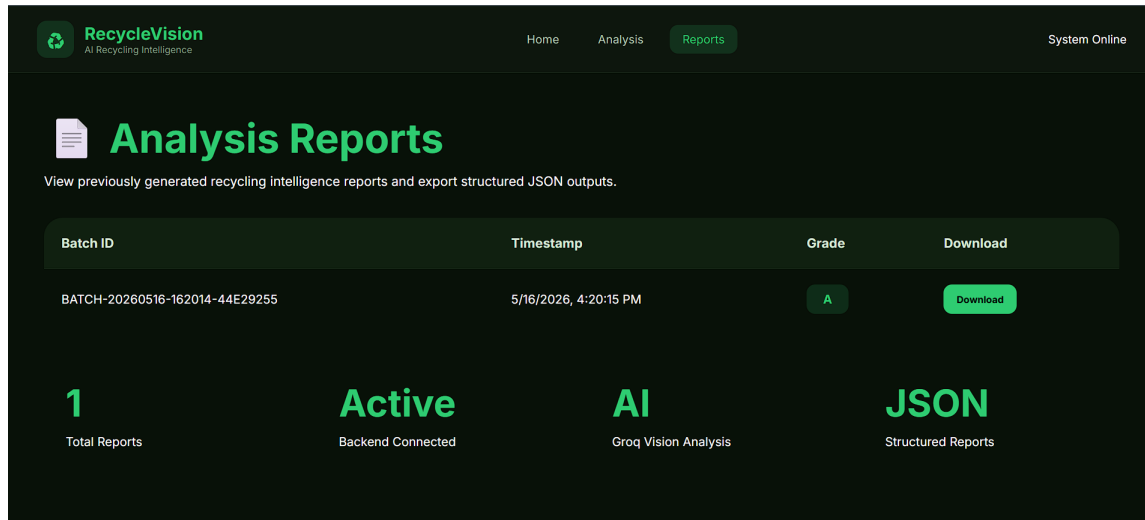


Figure 4: Reports Dashboard

6 Data Strategy

6.1 Current Prototype

The current implementation relies on pretrained multimodal AI models without a custom training dataset.

6.2 Future Data Collection Strategy

Future versions would collect:

- Recycling facility images
- Bottle composition examples
- PET and Non-PET samples
- Contaminated material images
- Multiple environmental conditions
- Diverse bottle shapes and colors

6.3 Data Labeling Strategy

Collected data would be labeled using:

- Human annotators
- Quality assurance verification
- Multi-review validation

- Active learning workflows

Labels would include:

- Bottle count
- PET classification
- Contamination regions
- Quality grades
- Color distribution

6.4 Continuous Improvement

The system could improve over time using:

- Operator feedback loops
- Human review corrections
- Incremental retraining
- Active learning pipelines
- Model performance monitoring

7 Evaluation Plan

Accuracy alone is insufficient for evaluating operational recycling systems.

The following business-focused metrics are important:

7.1 Bottle Count Error

Measures difference between predicted and actual bottle counts.

7.2 Confidence Calibration

Evaluates whether prediction confidence matches actual reliability.

7.3 False Contamination Alerts

Measures incorrect contamination warnings which may reduce operational efficiency.

7.4 Human Review Rate

Tracks percentage of batches requiring manual verification.

7.5 Inference Latency

Measures response time for operational usability.

7.6 Operational Throughput

Measures number of batches processed per unit time.

7.7 Cost Savings

Evaluates reduction in manual inspection effort and operational inefficiency.

8 MLOps and Production Strategy

8.1 Current Prototype

Current prototype characteristics:

- Single FastAPI backend
- Local storage / SQLite persistence
- Browser-based report storage
- Manual deployment pipeline

8.2 Production Improvements

A production-grade system would include:

- Cloud database infrastructure
- Scalable object storage
- CI/CD pipelines
- Kubernetes orchestration
- Model monitoring
- Logging and observability
- GPU-based scalable inference
- Model versioning
- Batch processing queues

8.3 Monitoring Strategy

Production monitoring would track:

- Prediction drift
- System latency
- Infrastructure utilization
- Confidence degradation
- Failure rates

9 Limitations

The prototype has several limitations:

- Bottle counting is estimation-based
- No dedicated object detection model
- Results depend on multimodal reasoning
- Limited real-world recycling dataset
- Free-tier deployment infrastructure limitations
- Local storage used for report persistence

These limitations were accepted intentionally to prioritize architectural exploration and rapid prototyping.

10 Future Improvements

Potential future enhancements include:

- YOLO-based object detection
- Image segmentation models
- OCR for label extraction
- Historical analytics dashboard
- Real-time video processing
- Cloud-native infrastructure
- Authentication system
- Large-scale dataset training
- Automated retraining pipelines

11 Conclusion

RecycleVision demonstrates a practical AI-powered recycling intelligence workflow using modern multimodal AI systems and full-stack deployment architecture.

The project successfully integrates:

- React frontend dashboard
- FastAPI backend API
- Groq Vision AI
- Cloud deployment
- Structured operational reporting

The prototype prioritizes architectural thinking, practical trade-offs, and rapid implementation while clearly acknowledging limitations and future production requirements.

Thank You